

DEEPAK KUMAR PATEL

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Summary

Machine Learning and Generative AI enthusiast with hands-on experience in building end-to-end machine learning systems, real-time REST APIs, and deep learning models. Strong background in Python, data structures, and algorithms, with applied experience in fraud detection, medical image analysis, and retrieval-augmented generation (RAG) systems.

Education

Indian Institute of Information Technology Bhagalpur

Nov 2022 – July 2026

Bachelor of Technology in Computer Science and Engineering

Bhagalpur, Bihar

Grade: 7.71 (overall)

Skills

Languages: C++, Python, SQL, HTML, CSS

Machine Learning & AI: Machine Learning, Deep Learning, CNNs, Logistic Regression, Feature Engineering, Model Evaluation

Tools & Frameworks: TensorFlow, NumPy, Pandas, Scikit-Learn, FastAPI, Flask, LangChain, FAISS, OpenAI API

CS Fundamentals: Data Structures, Algorithms, OOPs, DBMS, Operating Systems, REST APIs, Postman

Projects & Experience

Real-Time Fraud Detection System ([GitHub Link](#))

Machine Learning — Python, Scikit-learn, FastAPI, Pandas

Designed and deployed an end-to-end real-time fraud detection pipeline for financial transactions under extreme class imbalance.

- Processed and analyzed **6M+ transaction records**, performing EDA and engineering **15+ real-time-safe behavioral features** while preventing data leakage.
- Trained a class-weighted Logistic Regression model achieving **95%+ fraud recall** and **0.92 ROC-AUC**, prioritizing risk reduction over raw accuracy.
- Implemented probability-based thresholds for **BLOCK / REVIEW / ALLOW**, reducing false negatives by **30%**.
- Built automated data validation checks that improved input reliability by **20%** and reduced model retraining frequency by **25%**.

PDF Chat Application ([GitHub Link](#))

Generative AI — Python, LangChain, Streamlit, FAISS, OpenAI API

Developed a retrieval-augmented generative AI system for conversational querying across large PDF document collections.

- Implemented end-to-end PDF ingestion, chunking, and embedding pipeline for **1000+ pages** using FAISS, reducing query latency by **40%**.

Disease Detection Model (Pneumonia, COVID, TB) ([GitHub Link](#))

Deep Learning — Python, TensorFlow, CNN, OpenCV

Built a deep learning-based medical imaging system to classify chest X-ray images into multiple disease categories.

- Preprocessed and trained on **5,000+ chest X-ray images**, developing a multi-class CNN achieving **85%+ classification accuracy**.
- Applied data augmentation techniques that improved model generalization and reduced overfitting by **18%**.

Certifications

Supervised Machine Learning: Regression and Classification [Coursera](#) / [Andrew Ng](#)

- Learned linear/logistic regression, gradient descent, and evaluation techniques.

Advanced Learning Algorithms [Coursera](#) / [Andrew Ng](#)

- Covered neural networks, regularization, optimization, and decision trees.

Unsupervised Learning, Recommenders, RL [Coursera](#) / [Andrew Ng](#)

- Studied clustering, anomaly detection, recommender systems, and reinforcement learning basics.

Achievements & Profiles

- Achieved Knight Badge on LeetCode (**peak rating: 1860**) by solving **500+** data structures and algorithms problems. [LeetCode](#)
- Earned **2-Star** rating on CodeChef demonstrating strong problem-solving skills. [Codechef](#)
- Secured top **160 rank** among 2000+ students in GeeksforGeeks institute rank list. [GeeksforGeeks](#)